

Problem H. String Problem

Time Limit 2000 ms

Mem Limit 262144 kB

Input File stdin

Output File stdout

Boy Valera likes strings. And even more he likes them, when they are identical. That's why in his spare time Valera plays the following game. He takes any two strings, consisting of lower case Latin letters, and tries to make them identical. According to the game rules, with each move Valera can change **one** arbitrary character A_i in one of the strings into arbitrary character B_i , but he has to pay for every move a particular sum of money, equal to W_i . He is allowed to make as many moves as he needs. Since Valera is a very economical boy and never wastes his money, he asked you, an experienced programmer, to help him answer the question: what minimum amount of money should Valera have to get identical strings.

Input

The first input line contains two initial non-empty strings S and t , consisting of lower case Latin letters. The length of each string doesn't exceed 10^5 . The following line contains integer n ($0 \leq n \leq 500$) — amount of possible changings. Then follow n lines, each containing characters A_i and B_i (lower case Latin letters) and integer W_i ($0 \leq W_i \leq 100$), saying that it's allowed to change character A_i into character B_i in any of the strings and spend sum of money W_i .

Output

If the answer exists, output the answer to the problem, and the resulting string. Otherwise output - 1 in the only line. If the answer is not unique, output any.

Examples

Input	Output
uayd uxxd 3 a x 8 x y 13 d c 3	21 uxyd

Input	Output
a b 3 a b 2 a b 3 b a 5	2 b

Input	Output
abc ab 6 a b 4 a b 7 b a 8 c b 11 c a 3 a c 0	-1